Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (`uXXXXXXX@anu.edu.au`); PE supports ANU SSO.

   https://pollev.com/anuiml2026



Photographing or recording the quiz questions is not permitted

Lecture 4 - Quiz 1 (120 seconds)

L4Q1 - A bank fits a logistic regression and denies credit when the estimated probability of default exceeds $\frac{1}{2}$. A student objects: *"the perceptron of Lecture 1 can only draw a straight line, whereas logistic regression fits an S-shaped curve, so it is the strictly more powerful model"*. What is the correct assessment?

- A) The student is wrong for this bank, but right in general: it is the threshold $\frac{1}{2}$ that makes the boundary a hyperplane. With a threshold reflecting what a default costs the bank, say 0.15, the S-shape is cut at another height and the boundary curves.
- B) The student is wrong at any threshold: the rule compares the score with a constant, so the boundary is a hyperplane and the set of classifiers considered is exactly the perceptron's.
- C) The student is wrong, and more strongly than they realise: the two procedures share the hypothesis set *and* the answer, since both return the hyperplane with the smallest E_{in}.
- D) The student is right, though not for the reason given: the perceptron's hypothesis set consists of classifiers, while the logistic model's consists of functions $\eta(\cdot \mid \boldsymbol{\theta})$ with values in $[0, 1]$, a far larger set, and its extra members produce decision boundaries no hyperplane can reproduce.

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- A) The student is wrong for this bank, but right in general: it is the threshold $\frac{1}{2}$ that makes the boundary a hyperplane. With a threshold reflecting what a default costs the bank, say 0.15, the S-shape is cut at another height and the boundary curves.
- **B) The student is wrong at any threshold: the rule compares the score with a constant, so the boundary is a hyperplane and the set of classifiers considered is exactly the perceptron's.** *What it supplies is a criterion for choosing among those hyperplanes, maximum likelihood, that is the cross-entropy loss, and a probability for each applicant. Moving the threshold to c only shifts the intercept: $\{\hat{\eta} > c\}$ is a half-space for every c.*
- C) The student is wrong, and more strongly than they realise: the two procedures share the hypothesis set *and* the answer, since both return the hyperplane with the smallest E_{in}.
- D) The student is right, though not for the reason given: the perceptron's hypothesis set consists of classifiers, while the logistic model's consists of functions $\eta(\cdot \mid \theta)$ with values in $[0, 1]$, a far larger set, and its extra members produce decision boundaries no hyperplane can reproduce.

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Lecture 4 - Quiz 2 (120 seconds)

L4Q4 - The lecture states that **fully-grown trees should never be used**, and then defines a random forest as a large collection of **fully-grown** trees. A student sees a contradiction. What is the right reply?

- A) There is no contradiction: each tree is grown on a bootstrap sample, which contains only about two thirds of the distinct observations, so the trees are grown on less data than is available and none of them is fully grown in the sense the warning had in mind.
- B) There is no contradiction: at every split a tree may only consider a different random subset of the variables, so each tree can no longer isolate every observation in a leaf of its own, and restricting the model is precisely what the warning asked for.
- C) There is no contradiction: a fully-grown tree has low bias and very high variance, and it is the variance that the vote in a random forest removes.
- D) The student is right: a forest of fully-grown trees attains an in-sample error of essentially zero as well, and it overfits exactly as much as one such tree.

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- B) There is no contradiction: at every split a tree may only consider a different random subset of the variables, so each tree can no longer isolate every observation in a leaf of its own, and restricting the model is precisely what the warning asked for.
- **C) There is no contradiction: a fully-grown tree has low bias and very high variance, and it is the variance that the vote in a random forest removes.** *The bias is left where it was, while the disagreements between the trees, made possible by the bootstrap and by the random subset of variables considered, cancel out. Individually they overfit; collectively they do not. The price of the vote is the one thing a single tree was best at: interpretability.*
- D) The student is right: a forest of fully-grown trees attains an in-sample error of essentially zero as well, and it overfits exactly as much as one such tree.

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